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An Interpretable ML-Based Framework for Credit Card Fraud Detection Using Ensemble Learning and Feature Selection

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***Abstract*—One of the most crucial problems in the financial sector is fraud detection in a credit card, where rogue transactions make a huge risk for both financial institutions and customers. Addressing fraud detection in the right way is key to secure payment systems, even more so because these systems minimize financial loss for consumers and market confidence. This paper explores various machine learning algorithms to achieve utmost accuracy by using a balanced, no-noise dataset, for detecting fraudulent credit card transactions. The analysis starts with preprocessing of the data, followed by feature selection (FS) using techniques like selectkbest with chi-square, XGBoost, and mutual information. Then trained different models like XGBoost, LGBM, GBM, Random Forest (RF), KNN, Logistic Regression (LR), Decision Trees (DT) and showed how RF, KNN, XGBoost were better than most. To boost the performance of the model more, a stacking ensemble technique combining RF and KNN was applied and it reached to 99.99% accuracy. Model performance was evaluated through metrics including accuracy, precision, recall, f1 score and ROC-AUC. LIME is used for model explainability.**

Keywords—credit card, accuracy, feature importance, fraudulent transaction, machine learning, lime

I. INTRODUCTION

Information and internet technologies have tremendously changed the business world including the financial one. The COVID-19 pandemic has accelerated this digital trend, boosting e-commerce growth and the significance of secure online payments [1]. The extensive adoption of credit cards provide a target for fraudsters among this massive audience, especially since a high percentage of worldwide transactions are done by card payment. Research indicates about 0.05% of all monthly active accounts are not genuine, 5 in every 10,000 active accounts on average are fake [2]. A thorough investigation by the Association of Certified Fraud Examiners (ACFE) found that fraud can be expensive [3]. Fraud causes more than $4.7 trillion in annual losses around the world. In its analysis of 2,110 cases in 133 geographies, this loss was a jaw-dropping $3.6 billion, an average of $1.78 million for each case [3]. Obtained via an organization of more than 90,000 anti-fraud experts, the information illustrates the prevalence of payment fraud across a wide range of industries [4]. Banking was done only face-to-face before 1996, when the United States' first online banking program was made available by Citibank and Wells Fargo Bank in 1996 [5]. Global credit card fraud losses reached USD 33.45 billion in 2022 and envisaged to exceed 43 billion US dollars by 2028, as per the Nilson Report [6]. One of the largest credit card frauds in UK history resulted in total losses of GBP 17 million. Notably, this took place in mid-2000s where a network of international fraudsters worked together to steal personal information from more than 32,000 credit cards [7]. The trend of using Artificial intelligence (AI) methods bringing in data mining and Machine Learning (ML), fraudulent behavior in the sector of finance has been discovered [8,9]. In the more recent past, different ML techniques have been employed in order to identify credit card fraud [10]. Computational efficiency is a key factor to enable most ensemble machine learning models to scale and be useful in practice. Ensemble can handle complex algorithms, sophisticated feature engineering and the composition of multiple base models without losing speed or responsiveness because of computational efficiency.

Traditional and expert-based fraud detection systems are expensive and difficult to maintain, while modern ML models suffer from lack of transparency, leading to regulatory and ethical challenges [11]. To tackle these problems, this research offers the interpretability trait of methods like ensemble of machine learning combining RF, Boosting, KNN with feature selection techniques as decision

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formulation stays clear. The approach has the objective of minimizing false alarms, as well as missed frauds to provide financial institutions with a more economically viable and regulatory compliant solution. This paper presents the following key work:

- The model integrates several machine learning classifiers such as XGBoost, light gradient boosting (LGBM), gradient boosting (LGBM), k-nearest neighbor (KNN), decision tree (DT), random forest (RF) and linear regression (LR). The best results were obtained for Random Forest (RF) with an accuracy of 99.98%.

- To improve accuracy, The stacking ensemble method is used combining best performing algorithms. Highest performance (99.99% accuracy) was offered from the RF+KNN with 99.97% precision, 100% recall and 99.99% f1-score respectively. ROC curve & confusion matrix visualization have been employed to understand more about its predicting ability.

- Feature importance is utilized to enable model understanding and optimization by providing insight into which features are most relevant to fraud detection and ensured the best features are leveraged for improved accuracy.

- The utilization of LIME (Local Interpretable Model-Agnostic Explanations) with the best model provided better interpretability and transparent insights into decision making which is important for fraud detection applications.

II. Literature Review

Ranging from a naive rule-based system to more sophisticated methods incorporating deep learning and machine learning techniques, the literature on credit card fraud detection illustrates a spectrum of approaches.

Soleymanzadeh et al. [12] investigates the important problem of cyber-attack and fraud detection in IoT networks. In this work, they suggest a stacked ensemble, using multiple ML models to enhance the detection accuracy. They experiment using three datasets which are: credit card fraud, NSL-KDD and UNSW-NB15. Stacked ensemble model shows better performance which is 93.5% in credit card fraud detection (CCFD) and 95.15% for cyber-attack detection respectively than individual ones. The model has limitations, including its greater computational power requirement and its possible overfitting for larger datasets.

Bin Sulaiman and et al. [13] addresses the question of CCFD, suggesting a hybrid model integrating Artificial Neural Networks (ANN) and Federated Learning (FL) for enhancing privacy in fraud detection. Their proposed FL-ANN model achieved an accuracy of 99.96% in the CCFD. But this approach introduces barriers such as different financial institution regulation and possible security risks about sharing the trained models.

Obukohwo et al. [14] proposed the Random Forest (RF) Classifier enhanced by Synthetic Minority Over- sampling Technique (SMOTE) for CCFD to handle the dataset's class imbalance problem. The model was tested in Kaggle credit card fraud dataset. RF performed the best with 99.19% accuracy after the application of feature selection using chi-square and SMOTE outperformed all other models.

Xie et al. [15] presented HELMDD, a hybrid ensemble learning model with resampling method (RMDD) to handle class imbalance in credit card fraud detection. It has been tested over the Kaggle and a private bank dataset, achieving an 0.9853 AUC and 0.8623 savings rate in the Kaggle dataset, and 0.7941 AUC and 0.6696 savings rate in a private dataset. This outperforms other models in the detection of both fraudulent and non-fraudulent transactions.

Singh et al. [16] address the class imbalance problem in credit card datasets with anomaly detection for fraud detection. They implemented two unsupervised algorithms-Isolation Forest (IF) and the Local Outlier Factor (LOF) and improved their performance using different resampling techniques including Random Undersampling, AllKNN, SMOTE and SMOTE-ENN.

Jemai et al. [17] use ensemble learning to analyze several classifiers such as Naive Bayes, RF and XGBoost in two datasets: a real-world one obtained from European card holders and synthetic. The work compares the classifiers' performance under various data sampling schemes such as SMOTE, in order to enhance fraud detection accuracy.

Unlike many previous studies, this study is utilized by a balanced and cleaner dataset which is a real-world dataset supplemented with synthetic data without any external balancing techniques such as SMOTE or RUS. However, with feature importance-based selection this was much improved and we reached nearly-perfect accuracy and recall.

III. Materials And Methods

Fig. 1 showcases the complete methodology of the proposed work starts from data preprocessing where min-max scaling is applied to normalize the dataset, a split of dataset is taken where 80% are used for training and 20% for testing. Mutual information, selectkbest with chi-square and XGBoost evaluation of feature importance is done after splitting. Then seven ML algorithms which are XGBoost, LGBM, GBM, RF, LR, KNN, and DT are applied to the data.

To improve model accuracy, RF and KNN are bundled in an ensemble technique, stacking. LIME is applied to pinpoint the greatest features responsible in order to predict fraudulent or non-fraudulent cases.

A. Dataset

The dataset contains a record of European credit card transactions in 2023 with over 550,000 records which was obtained from Kaggle [18]. It is a balanced and clean dataset that has real-world records supplementing with synthetic records. Table I provides an overview of the dataset. This has 31 feature columns that can be broken into three categories. Anonymized features (V1 to V28) and "Amount" columns are of float type, whereas "id" and "Class" are of integer type. The dataset has no missing values, which is introducing the dataset is complete and we can proceed to modeling. The data is also fairly equal in its fraud/non-fraud composition which gives a great base for training effective fraud detection models.

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TABLE I. DATASET DESCRIPTION

Feature Name	Feature Description	Type of Data
id	Transaction key for each transaction	Integer
V1 - V28	28 anonymized features based on factors related to the transaction such as transaction location, time etc.	Float
Amount	The monetary amount of money that was exchanged	Float
Class	The target variable: 1 state a fraudulent transaction, 0 states a non-fraudulent transaction	Integer

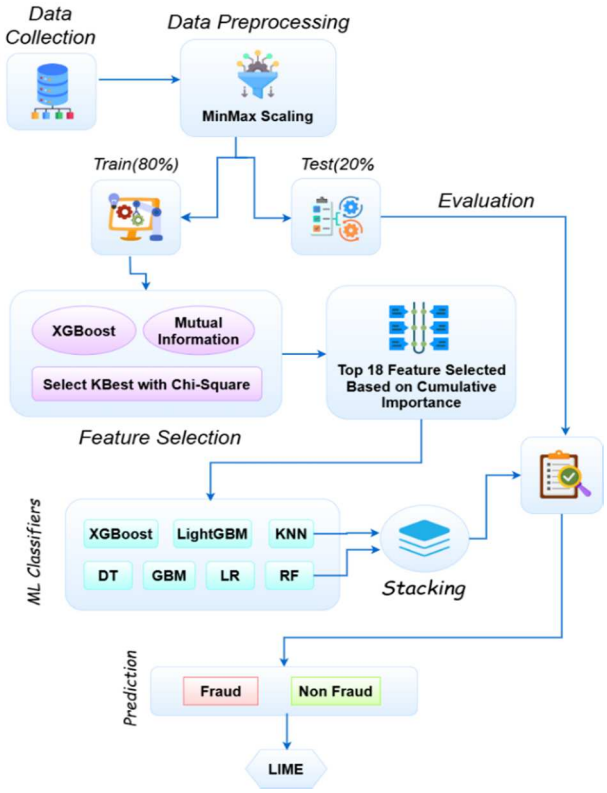


Fig. 1. Credit card fraud detection model

B. Data Preprocessing

To prepare it for machine learning, the dataset was purified by eliminating the "id" column first because the column has zero contribution to the prediction. Fig. 2 shows the correlation heatmap which depicts the strength and direction of relationship between features, with yellow indicating strong positive correlation and blue weaker or negative correlation, significant to identify which features are most influential in getting good model performance. Most of the anonymized features (V1 to V28) exhibit strong positive correlations and the feature "Amount" is showing weak correlations with other features. The "Class" variable depicting whether transactions is fraudulent or not is similar to specific anonymized features (like V10, V13), but has been observed to be a bit correlated. The bar plot in Fig. 3 represents the proportion of class 1 (fraudulent) and class 0 (non-fraudulent) transactions in the dataset. The number of instances for both classes is showing that the fraction between fraudulent and non-fraudulent transaction is balanced.

Min-Max Scaling- Min-Max scaling is applied to normalize the features in a fixed range from 0 to 1. Chi-

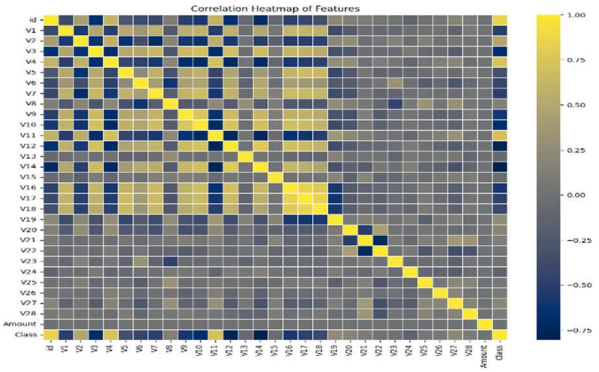


Fig. 2. Correlation heatmap of the dataset.

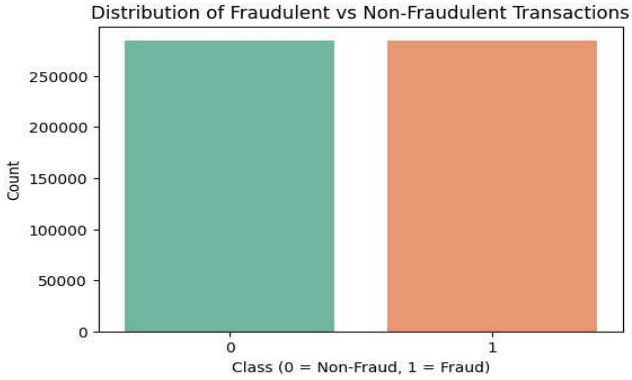


Fig. 3. Bar chart of dataset label distribution

Square FS method will be needed for the feature importance and it strictly needs non-negative values. As the values of features V1 to V28 had a negative part, therefore scaling was a most important step to transform all values into the range of non-negative. It was also crucial for other algorithms to select features that mainly work with non-negative values.

Data Splitting- After that, the data was divided into training and testing sets, with 20% going toward testing and 80% going toward training. This warrants the model will learn from most of the data and will be tested on a different, unseen, piece of data during its training.

Feature Importance- For feature importance, we have selected features using 3 methods – mutual information, selectkbest with chi-square, XGBoost and merges the output to select 18 important features based on their cumulative significance. The mutual information score assesses the dependency of a feature and the target variable and it sorts the features according to their influence in predicting the target. Chi-square is a statistical method to select the input features where random data is chosen based on two variables and the independent variables are obtained from huge data samples [19]. Selectkbest with chi-square determines whether a specific feature is statistically independent of the target variable and the features that seem to be associated are selected. The feature importance using XGBoost is calculated based on how the model has contributed towards prediction. Then all the three methods have been taken and the importance was calculated. The final 18 features were selected using a hybrid ranking approach that aggregated normalized importance scores from mutual information, chi-square, and

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XGBoost methods, retaining features contributing to 90% of the total cumulative importance to ensure a systematic and data-driven selection. The selected features are plotted using

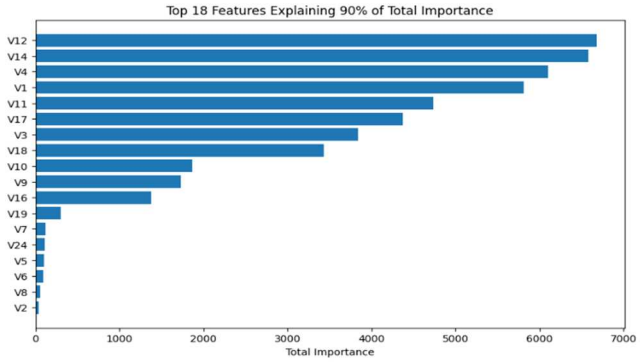


Fig. 4. To 18 selected features

a horizontal bar chart in terms of their total importance for creating a fraud detection model which is shown in Fig. 4 The figure visualizes the important features that were obtained by the three feature importance techniques.

C. Machine Learning Algorithms

Random Forest- It is a component of an ensemble learning technique that uses several decision trees to generate predictions in order to upgrade prediction accuracy. This decision tree is based on the idea of an n-ary tree that solves classification and regression tasks [20]. Creating each decision tree in the forest is a key ingredient for Random Forest to work. Random Forest can be expressed by equation (1).

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i(x) \tag{1}$$

where $\hat{y}$ is the final prediction, N is the number of trees and $T_i(x)$ is the prediction for the i-th tree given input x.

IV. Result And Discussion

The performance of different ML models with feature selection in terms of accuracy, precision, recall and f1 score has been summarized in Table II. In this study, GBM was implemented using the GradientBoostingClassifier from Scikit-learn, while LGBM was implemented using the LGBMClassifier from the Microsoft LightGBM library.

TABLE II. Comparison of ML and Ensemble Models

Machine Learning Models				
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
XGBoost	0.9995	0.9989	1.0000	0.9995
LGBM	0.9984	0.9969	0.9998	0. 9984
GBM	0.9777	0.9880	0.9672	0. 9775
DT	0.9977	0.9969	0.9985	0. 9977
RF	0.9998	0.9997	1.0000	0. 9998
KNN	0.9987	0.9973	1.0000	0. 9987
LR	0.9616	0.9821	0.9403	0. 9607
Ensemble Machine Learning Model (Stacking)				
RF+ XGBoost	0.9998	0.9997	1.0000	0.9998
XGBoost+KNN	0. 9998	0.9996	1.0000	0.9998
RF+KNN	0.9999	0.9997	1.0000	0. 9999

$$Acuracy = \frac{TP+TN}{TP+FP+TN+FN} \tag{2}$$

$$Pecision = \frac{TP}{TP+FP} \tag{3}$$

$$Recall = \frac{TP}{TP+FN} \tag{4}$$

$$F1\ Score = 2\frac{Precision.Recall}{Precision+Recall} \tag{5}$$

These models were treated as distinct, trained and evaluated independently. Overall, Random Forest (RF) is the best performing model with the highest accuracy, precision, recall and f1 score and it is the most reliable model overall. Logistic Regression (LR) shows a poor result (96.16% accuracy) compared to all the other algorithms and it is particularly quite low on recall and F1 score, which suggests a poor classification ability. The metrics used to evaluate the algorithm performance, accuracy, precision, recall, and f1-score are calculated with the equations (2-5).

A confusion matrix compares actual labels predicted by a classification ML model and it is a top-level summary table of model performance. The confusion matrix in Fig 5 exhibits a great performance with 17 false positives and no false negatives, which means catching all the fraud and no one got missed. For the final result, XGBoost provided a great accuracy with 99.95%, RF provided 99.98%, KNN provided 99.87%. Then we combine RF+KNN, RF+XGBoost and XGBoost+KNN using another ensemble technique - stacking.

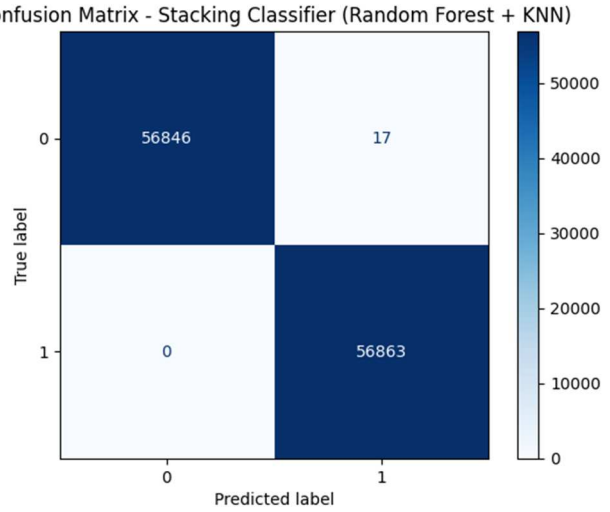


Fig. 5. Confusion matrix after applying stacking with RF+KNN

As seen from Table II, the stacked model RF+KNN further enhanced the performance to 99.99%, even though individual models like RF and KNN performed exceptionally well with accuracy scores of 99.98% and 99.87%, respectively. As a bagging-based ensemble of decision trees, RF is immune to overfitting and is great at learning the global pattern by selecting features randomly and voting majority. Conversely, KNN is a non-parametric method that is an instance based learner and can effectively characterize the local neighborhood structure. Their contrasting mechanisms, RF's generalization versus KNN's locality sensitivity offer diverse error profiles. By combining them, we exploit this diversity in order to reduce the overall prediction variance and promote robustness. This synergy is further evidenced by the promising results of the RF+KNN ensemble outperforming all individual which shows that combined performance is more than multipliable when it comes to two

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modalities. Additionally, stacking lessens the possibility of overfitting that can occur when using just one model.

ROC curve in Fig. 6 shows the performance of classification models as per the True Positive Rate vs False Positive Rate. AUC (Area Under the Curve) estimates how well the model will classify the data. The closer the value is to 1, the better the model's performance. There are perfect AUC values of 1.0000 for XGBoost and RF that signify perfect

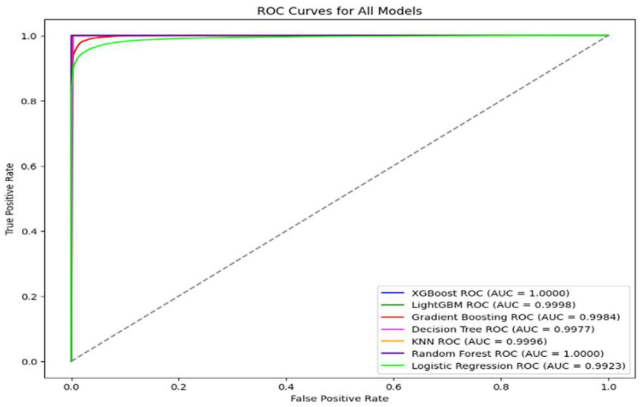


Fig. 6. ROC curve analysis of ML algorithms

classification. KNN has also an impressive AUC of 0.9996, but is a little less ideal because it can be sensitive to data noise and structure. However, in this case XGBoost and RF proved to be the most consistent.

This study compared multiple ML models under the influence of feature importance-based selection. The results highlight an interesting observation shown in Table III, as the ensemble models including XGBoost, LightGBM and RF

TABLE III. MODEL PERFORMANCES WITHOUT FEATURE SELECTION (FS)

Model	Accuracy	Precision	Recall	F1-Score
XGBoost	0.9998	0.9999	0.9996	0.9998
LGBM	0.9998	0.9999	0.9996	0.9998
GBM	0.9997	0.9998	0.9996	0.9997
DT	0.9996	0.9996	0.9996	0.9996
RF	0.9998	0.9999	0.9997	0.9998
KNN	0.9990	0.9998	0.9982	0.9990
LR	0.7924	0.7344	0.9162	0.8153
RF+KNN	0.9998	0.9999	0.9996	0.9998

kept near-complete performance with and without FS. The recall scores flatted above 0.999 nearly in every case. Simpler models such as LR and KNN exhibited dramatic betterment. The performance of LR improved significantly with FS, with accuracy increasing from 0.7924 to 0.9616 and f1-score increasing from 0.8153 to 0.9607. For KNN, perfect recall of 1.0000 was achieved, increased from 0.9982 after FS.

Improvements like these are particularly important in CCFD, where missing potential fraudulent transactions could lead to losses. According to the results, feature selection not only improves efficiency and interpretability but also improves the model's capabilities to make correct fraud predictions, particularly in models sensitive to irrelevant or noisy features. Fig. 7 depicts the LIME explanation of the stacking classifier (RF + KNN with LR), which portraits a logical, easy-to-understand rationale for predicting that a transaction is "Not Fraud" with 100% confidence. The most influential features toward the prediction of "Not Fraud" were V4 (0.32), and V1 (> 0.76), and toward prediction of "Fraud" were V14, V18, and V7 (all < 0.10). LIME holds that the stacking model uses relevant features consistently and interpretably, locally

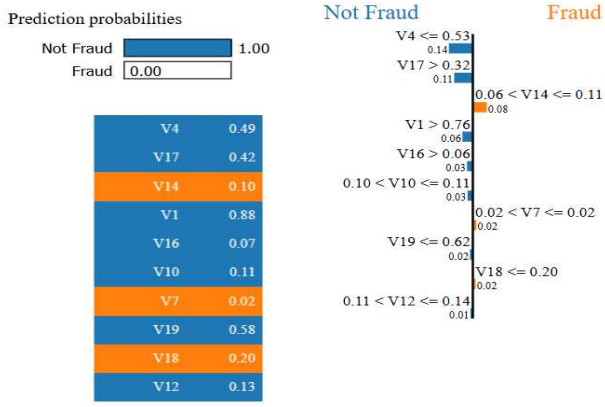


Fig. 7. LIME visualization of key features influencing fraud predictions

approximating the decision boundary and thereby improving model explainability to facilitate the use of high-stake prediction in domains such as fraud detection where accountability for decisions is essential.

TABLE IV. HYPERPARAMETERS USED IN ML MODELS

Model	Hyperparameters
XGBoost, LightGBM, GBM, Decision Tree, Random Forest	random_state=42
KNN	Default n_neighbors=5
Logistic Regression	random_state=42, max_iter=1000
Stacking Classifier	stack_method='predict_proba', final_estimator=LogisticRegression()

The hyperparameters utilized by the different models in the CCFD system are compiled in Table IV. While certain models, such as Logistic Regression, additionally specify extra parameters, like max_iter=1000 to restrict the number of iterations for convergence, the majority of models require random_state=42 to assure reproducibility of findings.

Table V highlights the exceptional performance of the proposed method. The RF+KNN stacking model has 99.97% precision score, recall of 100%, f1–score of 99.99% and produced an accuracy of 99.99%. Moreover, Singh and Jemai explored anomaly-based or shallow ensembles that were not yet recommended for deployment, unambiguous decision paths and scalability. Compared with other reference studies, our solution has significant progress in terms of technical performance and practical application in CCFD. Although most of the earlier works obtained high accuracy through ensemble or anomaly detection methods, they had little interpretability and were difficult to apply in reality. This work is unique due to its ability to achieve accuracy of 99.99 percent, considering a refined stacking ensemble of Random Forest and KNN, directed by a robust feature selection pipeline through mutual information, chi-square, and XGBoost. In contrast to previous work that omitted feature

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selection, or only employed simple filters, we demand dimensionality reduction for the purpose of enhanced generalization. Xie et al.'s HELMDD is not interpretable while our model also integrate LIME to provide transparency and interpretability in the model's decision which is an important

TABLE V. COMPARATIVE PERFORMANCE ANALYSIS OF PROPOSED MODEL WITH EXISTING CREDIT CARD FRAUD DETECTION MODELS

Author	Dataset	Method	Accuracy
Soleymanz adeh et al. [12]	Credit Card Fraud, NSL-KDD, UNSW-NB15	Stacked Ensemble - KNN, DT, NB, SVM, AB, GB, RF, ET, MLP, XGBoost	95.15%
Sulaiman et al. [13]	-	FL-ANN Model	99.96%
Aghware et al. [14]	Kaggle credit card fraud dataset	RF with SMOTE	99.19%
Xie et al. [15]	Kaggle, Private Bank dataset	HELMDD	-
Singh et al. [16]	European credit card fraud, German credit card fraud dataset	Isolation Forest (IF), Local Outlier Factor	99.81%
Jemai et al. [17]	European Union dataset, Synthetic dataset	RF, NB, XGBoost	-
Proposed Model	Credit card fraud detection dataset 2023 [18]	**Feature Selection and Stacking (RF+KNN)**	99.99%

gap in trust and regulatory compliance that we believe the others have overlooked. We employ an ensemble model to get a good trade-off between higher performance using a pre-balanced dataset combined with real and synthetic data, which is a contrast with Aghware et al. where they applied SMOTE using default RF settings.

V. CONCLUSION

As such, this study outlines a credit card fraud detection methodology that embeds feature importance-based selection strategies. After Min-Max scaling and usage of feature importance, we took only the top-18 features that contributed to 90% of the overall importance. A few different types of ML models such as XGBoost, KNN and Random Forest were used, along with a stacking ensemble model to improve performance. The best result of accuracy 99.99% was obtained after executing stacking with RF+KNN. With LIME, we were able to achieve better comprehensibility of fraud prediction features. Through both feature selection as well as balanced dataset which contains real-world data mixing with synthetic data, the proposed model surpassed traditional approaches by retaining high accuracy, recall and precision. It helps to bring efficiency and accessible solution, which can be employed by any financial institute for the betterment of fraud detection. We acknowledge that the use of synthetic data can introduce limitations in terms of model generalization. While synthetic samples help achieve better metric scores during training and evaluation, they may not fully reflect the nuanced behavior of fraud in real-world transactions. Therefore, although our models showed excellent performance including near-perfect accuracy and recall, these results should be interpreted with caution. Future work can be done working with dynamic real-world datasets, emphasis on combined approaches that can handle imbalanced data, and utilization of advanced features selection techniques. While promising, limitations include the use of an anonymized and balanced dataset, which might not accurately capture the complexity of the real world. The work shows that Fraud Detection can draw advantages from feature importance-based preprocessing and ensemble models.

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